

7 From an Experimental Technology to a Mainstream Product

Implications for Generative AI as an Artistic Medium

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7.1 Introduction

Artists have long embraced new technologies, and artificial intelligence (AI) is no exception. In the 1970s and 1980s, pioneers such as Harold Cohen and David Cope explored the creative potential of AI. In the 1990s and 2000s, artists like Luke DuBois, Sam Lavigne, and Sven König worked with natural language processing, pattern recognition, and computer vision (Grba, 2022). However, AI art remained niche until the rise of deep neural networks in the late 2010s, which broadened artistic engagement. A key driver of this shift was the introduction of generative adversarial networks (GANs) (Cetinic & She, 2021), developed by Goodfellow et al. (2014), which enabled the creation of synthetic images. Effectively staged projects such as “*The Next Rembrandt*” (2016) and “*Portrait of Edmond de Belamy*” (2018), which sold for nearly \$500,000 at Christie’s, demonstrated the artistic promise of GANs and attracted public interest. Beyond these headline projects, a growing number of artists began to incorporate AI into their practices, contributing to the institutional establishment of AI art (Zylinska, 2020). Tech firms such as Google and OpenAI supported this through residencies, workshops and conferences. As major art institutions began to exhibit, fund, and promote AI-based artworks, AI became part of the mainstream art world.

In contrast, the rise of accessible text-to-image models such as Midjourney and Stable Diffusion since 2022 has been met with scepticism rather than the earlier enthusiasm. This scepticism stems from a variety of concerns (Bird et al., 2023), including copyright infringement since these models are trained on copyrighted scraped without consent or compensation (Samuelson, 2023); fears of automation replacing creative labour, leading to reputational and economic harm (Jiang et al., 2023); and backlash against the flood of AI-generated content, which is often criticized for lacking quality, authenticity, and originality (Bevilacqua, 2025; Rogers, 2025). Beyond these issues, a less discussed but significant reason for scepticism is the potential unattractiveness of generative AI as an artistic medium. Rapid technological advances and optimizations have transformed generative AI from an experimental technology into a mainstream product. Paradoxically, this very success in terms

of optimization and profit may be diminishing its artistic appeal, especially among those at the forefront of its use: artists.

Pioneering artists such as Mario Klingemann (2023a) have criticized the “*predictable and boring*” one-click nature of current tools, while art critic Ben A. Davis (2023) has warned of the spread of “*anti-creativity*”. AI researcher Ahmed Elgammal (2023) went further, claiming that generative AI is destroying creativity and predicting that “*the era of AI art may soon be over*”. While there are numerous studies addressing the ethical, legal, and economic implications of generative AI for artists (Caramiaux et al., 2025; Jiang et al., 2023; Piskopani et al., 2023), little research has explored how this socio-technical shift affects artistic engagement. How does the transition from experimental technology to mainstream product shape artistic practice? Are generative models, with their technical affordances, still compelling as an artistic medium? This chapter addresses these questions by first providing a brief overview of ethnographic research on artistic engagement with AI prior to 2022. Secondly, based on these findings, the chapter considers the possible implications of recent socio-technical changes for artistic practice, outlining questions for future research.

7.2 Artistic Engagement with AI Before the Rise of Generative AI

In the second half of the 2010s, a growing number of artists began to integrate machine learning (ML) techniques into their creative practices. Since then, so-called AI art has become part of the mainstream art world as academia provides conceptual tools and legitimacy through conferences and publications, art institutions increase visibility and market value, and the tech industry supports selected artists through patronage (Shelby et al., 2024). However, few ethnographic studies explore the practices of early AI artists prior to the rise of commercial, large-scale generative models (Caramiaux & Fdili Alaoui, 2022; Ploin et al., 2022; Wingström et al., 2024). While Caramiaux and Fdili Alaoui (2022) and Ploin et al. (2022) interviewed internationally recognized artists to explore visual artists’ engagement with AI, Wingström et al. (2024) interviewed computer scientists and new media artists to explore how AI might redefine creativity itself. The following section outlines key aspects of artistic engagement with early generative AI models, drawing on these ethnographic studies.

7.2.1 *Demanding and Manual Practices vs. Automation and Efficiency*

Contrary to dominant narratives framing AI as a tool for automation and efficiency (Caramiaux et al., 2025), artists describe working with data-driven AI as physically and cognitively demanding (Caramiaux & Fdili Alaoui, 2022; Ploin et al., 2022). Their work involves time-intensive technical research, selecting and/or creating models, manipulating code, collecting and curating datasets, and contextualizing results (Ploin et al., 2022). Artistic

workflows often resemble what Caramiaux and Fdili Alaoui (2022) call *crafting*: a hands-on, iterative approach of coding, tweaking, training, and playing with the model. Far from being automated or streamlined, these processes are described by the artist as labour- and time-intensive, involving both high levels of cognitive engagement requiring technical skills and meticulous manual tasks (Caramiaux & Fdili Alaoui, 2022; Ploin et al., 2022).

7.2.2 *Exploration vs. Execution*

Artists often describe their practice with AI as experimentation, exploration, and play (Wingström et al., 2024). As Caramiaux and Fdili Alaoui (2022) note, artists experience the model's behaviour rather than having a theoretical understanding of its functionality. Their engagement with AI is a highly embodied experience (Ploin et al., 2022), characterized by playful and sometimes frustrating interactions with the model's behaviour (Wingström et al., 2024). This contrasts with the industrial and scientific demand for reliable, predefined and accurate outcomes—an expectation paradoxically hard to meet, as AI systems exemplify an attempt to “control the lack of control” (Esposito, 2022), exposing their inherent unpredictability and opacity even in precision-driven domains. While scientists are concerned with producing results that are recognized as knowledge, artists describe working with AI as a playful, spontaneous process (Wingström et al., 2024), and that they do not adhere to the values that dominate research, science, and business. (Caramiaux & Fdili Alaoui, 2022). For example, artist David Young, interviewed by Ploin et al. (2022, p. 24), shares that he trains the model with minimal data, not only to understand the machine's behaviour, but also “*in contrast to the notion of AI being used for business, scale, and efficiency*”. Thus, artists often see AI not as a tool for predetermined, reliable results, but as an exploration in which the process outweighs the result (Caramiaux & Fdili Alaoui, 2022; Ploin et al., 2022).

7.2.3 *Mistakes, Glitches, and Surprise vs. Perfection and Precision*

Through embodied experience (Ploin et al., 2022) and the process of *crafting* (Caramiaux & Fdili Alaoui, 2022), artists gain an understanding of how the machine works, which not only enables them to predict its behaviour and achieve desired outcomes, but also to “hack” it, i.e., manipulate or subvert it for creative purposes (Cetinic & She, 2021; Shelby et al., 2024). Many artists choose to “work the model against the grain” (Ploin et al., 2022, p. 24), using AI in unexpected ways to discover surprising properties of the system, but also errors and machine failures. This leads to a key aspect of artistic engagement with earlier generative models: pioneer AI artists embrace errors and glitches as a source of inspiration and creative opportunity.

Moreover, the artists' interest in machine failure and glitches goes beyond the possibility of new aesthetics, but also reflects a political stance—artists oppose the efficiency-driven, standardized engineering culture of AI, which

prioritizes performance metrics (Caramiaux & Fdili Alaoui, 2022). As Anna Riedler explains in her interview with Caramiaux and Fdili Alaoui (2022, p. 11):

It's those mistakes that I think a lot of artists are really interested in, and it's also those mistakes that software engineers, by-and-large working at companies, are trying to get rid of (...) it's always a race where software engineers are optimizing out the mistakes, and the artists who want to grab them, and use them, and explore them.

The embrace of AI “failure” by artists echoes earlier movements where glitches have long held aesthetic and political significance in media art (Betancourt, 2016).

7.2.4 Uncertainty and Unpredictability vs. Control and Prediction

Empirical research identified the nondeterministic nature of AI—characterized by unpredictability and uncertainty—as a key factor in its appeal as an artistic medium. It is the ability of AI to produce unexpected, unpredictable results that plays a crucial role in working with AI (Caramiaux & Fdili Alaoui, 2022). Artists consider unpredictability as a fundamental part of AI's creativity (Wingström et al., 2024) and as a form of expressivity (Caramiaux & Fdili Alaoui, 2022). Similarly, Ploin et al. (2022, p. 78) note that it is the generative capabilities of AI that make it particularly compelling for artists, and what is seen as a “step change” from previous artistic tools. Artist Anna Riedler, interviewed by Ploin et al. (2022, p. 35), states:

I do think it is a different paradigm because I think just the possibility for it to just kind of do something totally unexpected is much higher. The possibilities of it doing something unexpected within algorithmic art are always within certain bounds. The type of unexpectedness that you get from machine learning, it's just so wild, and so crazy, and just amazing. I think it is, fundamentally, a shift.

The downside of machine unpredictability, however, is that there is less control over the system. Working with AI involves balancing the excitement of unpredictability with the frustration of having less control over the process (Ploin et al., 2022). For example, the artist Robbie Barrat, interviewed by Ploin et al. (2022, p. 44), explains:

A lot of my work is about misinterpretation, and that can be a double-edged sword. It's cool when the network misinterprets things and gives you surprising results, but it's also very frustrating when you cannot get the exact image that you want. I feel like I have a lot less control over the exact thing that I'm making. It's not like I'm a painter where I can

just place a brushstroke wherever I please. I'm working with a larger system. If I had to describe working with algorithms, I'd have to say it's a split between frustration and nice surprise all the time.

Artistic engagement with AI can thus be described as a constant process of negotiation between chance, automation on the one hand, and control on the other, or as [Caramiaux and Fdili Alaoui \(2022, p. 11\)](#) put it, “finding a balance between desired unpredictability and unwanted uncontrollability”.

Ethnographic research shows that artistic engagement with early generative AI models was not adoption of a passive tool, but rather a labour-intensive and cognitively demanding practice similar to craft. Rather than using AI to execute pre-defined ideas, artists often use it to explore and play with the system, embracing glitches and errors as sources of expressiveness and creativity. This approach contrasts with scientific and industrial practice, positioning artistic use of AI as divergent from the norms of optimization, accuracy, and efficiency. A crucial aspect of this practice is the balance that artists strike between controlling the system and allowing it to surprise them—navigating a space where chance and calculation, control and desired unpredictability are constantly negotiated ([Dzhimova & Tigre Moura, 2024](#)).

7.3 The Shift from an Experimental Technology to a Mainstream Product

7.3.1 Interface Optimization

Perhaps one of the most significant recent changes in generative AI has been the introduction of prompt-based user interfaces, which simplified the complexity of AI models and made natural language the primary mode of interaction. Platforms such as RunwayML and Midjourney also introduced web-based interfaces and APIs, simplifying back-end infrastructure and lowering the barrier to entry for non-technical users. Previously, generative AI tools required artists to understand coding, ML frameworks, and hardware, making artistic engagement time-consuming and challenging. However, this process of crafting ([Caramiaux & Fdili Alaoui, 2022](#)) helped artists gain a deeper understanding of the machine's behaviour, allowing them to not only predict but also manipulate its behaviour. Although early generative tools remained black boxes due to their opaque operation, often producing unexpected results, there were still many levels at which the artist had control over the process. In contrast, web-based platforms seem to limit this control and make the process more opaque by encapsulating the complexity of AI models behind simple text prompts, drag-and-drop interfaces, visual dashboards, and integrations with other tools via APIs.

An example of this potential loss of control can be found in navigating the so-called latent space—an abstract space of features the model has learned. Early on, artists often worked with generative models like GANs

(Goodfellow et al., 2014), which lacked natural language interfaces and did not incorporate the model’s textual “interpretation” of latent space. Artists manually explored this space by altering latent vectors—the internal numerical coordinates shaping what the model generates. This hands-on process offered a degree of control over unpredictability and allowed the discovery of unknown or unexpected regions in latent space—novel visual forms. In many artworks, this exploration became not only a way to make latent space tangible but also a form of expression in itself (Grba, 2022).

In contrast, today’s models use text prompts to generate images by mapping text to a broader latent space and generating corresponding images—effectively “jumping” to regions aligned with the prompt’s semantics. While this shift has made generative tools more accessible, it has obscured the underlying mechanics. As a result, users have less direct control over the generation process and cannot directly experience and influence how the text is mapped to the image.

7.3.2 *Optimization of Model Architecture, Training Strategies, and Data Regimes*

Further technical advances in generative AI have been driven by the rise of diffusion models, which outperform GANs by producing high-fidelity images through a stable, iterative denoising process. Systems like *Stable Diffusion* and others have scaled this architecture using massive text-image datasets, enabling the generation of highly detailed and semantically rich images from textual prompts. At the same time, foundation models trained on diverse multimodal data—often using self-supervised objectives—have reduced the need for task-specific tuning, enabling flexible control over aesthetic styles through natural language prompts.

These technical advances have significantly improved the fidelity, coherence, and semantic precision of the output. Accordingly, contemporary diffusion-based systems can produce polished, high-quality results driven by the prompt, dramatically reducing the errors and unpredictable outcomes typical of earlier models, such as GANs. Training on large datasets and using refined loss functions contributes to greater semantic fidelity, meaning the image accurately reflects the user’s intent as expressed in the prompt. As these systems are very good at following the rules, i.e., the instructions in the prompt, users get the results they expect. Any deviations, randomness, or surprises are “limited to what variations of our idea we might get” (Elgammal, 2023). However, as previously mentioned, pioneering artists view unpredictability as a key source of AI’s expressiveness and creativity. It seems that the optimization processes that have taken place in recent years have tamed the unpredictability of AI systems—the very characteristic that made AI attractive to artists as an artistic medium.

In addition, contemporary generative models are trained on large-scale datasets containing billions of web-scraped images, where dominant visual

conventions prevail. Consequently, these models internalize and reproduce mainstream and stereotypical aesthetic biases from their training data. As they extract general patterns, they favour common over rare and unique aesthetic styles (Manovich, 2002). Rather than embracing imperfections, accidents, or glitches, the model steers output toward the most statistically probable result, aligned with the dominant traits of its dataset. This tendency narrows the space of possibilities to the most “banana-like” or “strawberry-like” forms—that is, outputs corresponding to the densest clusters in the model’s latent space, where the most statistically frequent and visually prototypical representations of categories like “banana” or “strawberry” reside. These high-probability regions yield idealized, generalized versions learned from the training data—perfect simulations approaching what Baudrillard (1984) termed the *hyperreal*. Such an aesthetic default challenges the notion that AI is a neutral tool passively executing instructions. Instead, the model “interprets” prompts through a lens shaped by learned preferences, consistently leaning toward polished, idealized visuals. As artist Mario Klingemann (2023b) observes, “*certain ‘pretty’ aesthetics is rather difficult to escape in the streamlined and guard-railed creation process, which again gives the world the impression that AI art has to always look this way*”.

As a result of all these optimization processes, it seems that the “balance between desired unpredictability and unwanted uncontrollability” (Caramiaux & Fdili Alaoui, 2022, p. 11), which has been described as crucial for artistic engagement with earlier generative models, has been disrupted. On the one hand, the desired unpredictability is controlled, tamed, and stabilized by advanced model architectures, training strategies, and data regimes that minimize errors and deviations and enforce mainstream aesthetics by default. On the other hand, control over the generative process is reduced by encapsulating the complex inner workings of AI models behind simple and accessible but opaque interfaces—i.e., the unwanted uncontrollability is more likely. As Bredies (2014) notes in relation to the design of technologies in general, it seems that the same applies to the development of AI in particular, namely that “reducing complexity through simple or modular interfaces inevitably involves limiting the range of possible actions to those that are expected” (p. 74).

Accordingly, contemporary AI systems have moved from a nascent, unstable technology that allows for exploration and experimentation, to a stable and accurate working product that is designed and inscribed (Akrich, 1992) to be used in certain expected ways for certain expected purposes.

7.3.3 *Mass Adoption*

The technological advances as well as lower access costs and financial barriers, public attention to AI in the media, and investments in research and development, have led to rapid mass adoption of generative AI (Ali et al., 2025; Bick et al., 2024). Thus, generative AI has rapidly evolved from a niche technology used by pioneers to a mainstream technology used by the general

public. This mass adoption may further limit the appeal of generative AI as an artistic medium, as many artists have reported that it was the novelty and niche nature of the technology that fascinated them. For example, the artist Nicolas Boillot, interviewed by [Ploin et al. \(2022, p. 18\)](#), explains:

I first wanted to use machine learning because there was hype. There was huge hype. You want to be the first to use it. Every time a new tool is created, artists engage with it in their practice.

The artist Helena Sarin interviewed by [Ploin et al. \(2022, p. 18\)](#), argues similarly:

Technology drives the artistic process - because novelty is the thing. You need to invent or find breakthroughs, so you look around. You see a new medium and try to use it for your own purposes.

In addition, the mass adoption of generative AI has led to a flood of AI-generated content, resulting not only in saturation, but also in so-called “AI slop”—low-quality AI-generated content that lacks originality, effort, or purpose and is primarily fed into social media feeds to generate clicks ([Field, 2025](#)). Accordingly, generative AI seems to be losing not only its novelty, but also its value.

7.4 Conclusion and Future Research Agenda

Technologies that evolve from a mere new idea into an innovation and become established in societies go through a process that begins with an experimental technology used by a few—pioneers and early adopters—and ends with a mainstream product used by the majority ([Rogers, 2003](#)). This is also true of generative AI, which for a variety of reasons has moved rapidly from an experimental, nascent technology used by a few to a stable, mainstream tool used and adopted by many ([Bick et al., 2024](#)). Once defined by uncertainty, unexpected behaviour, and error, today’s AI systems are increasingly reliable, powerful, and sophisticated. This chapter explores the potential implications of this shift from a nascent and niche technology to an efficiently functioning commercial tool for artistic practice with AI. After briefly outlining the socio-technical changes and optimization processes that have taken place in the recent years, and drawing on empirical research into earlier artistic engagement with, the chapter argues that contemporary generative AI may be losing its appeal as an artistic medium.

The reasons for this loss seem to lie not only in the fact that generative AI is no longer a new technology with the potential to discover new aesthetics and visual languages, but above all in the fact that generative AI is no longer a site of unpredictability, surprise, and expression, but of “perfection”, scale, and efficiency. For many early adopters in the art world, AI was never about

perfect results or seamless automation; rather, they were fascinated by the unpredictability of early generative systems. The paradox is that as generative AI becomes more accurate or optimized, it seems to simultaneously limit the possibility of creating or opening up new possibilities. In other words, as the “*control of the lack of control*” over generative AI is value-optimized, it seems that the possibility of opening up new possibilities is simultaneously limited. This tendency toward “*perfection*” not only reinforces a certain homogeneity, but also suggests an absence of the friction necessary for innovative creation—the kind of friction that gives rise to new art forms and expressions. These conceptual considerations, however, require further empirical research. Accordingly, the following questions seem crucial for future research on the role of AI in artistic practice.

RQ1: *How do artists experience the socio-technical changes around generative AI?*

The transition of generative AI from a niche technology to a mainstream commercial product occurred at an unprecedented rate. Two months after its release, ChatGPT has reached 100 million users, making it the fastest growing consumer application in history (Milmo, 2023). However, there is limited ethnographic research on how this shift affects artistic engagement with AI. Accordingly, further empirical research is needed to explore how artists engage with the latest socio-technical developments and the extent to which they perceive and experience that AI has moved from a medium of exploration to a tool that performs and fulfils a function (Manovich, 2002). If so, how are artists adapting their approaches to maintain a sense of control, uniqueness and unpredictability? This in turn raises further questions: is there a difference in experience and coping strategies between pioneers, i.e. artists working with AI before mass adoption, and newcomers? Is there a difference between artists for hire and independent artists?

RQ2: *How do these socio-technical changes affect the sub-world AI art?*

Historically, artistic interest in new technologies often follows a familiar trajectory: exploration, appropriation, commodification, obsolescence. The early avant-gardes embraced new technologies not because they were established, but because they were in flux - poorly understood, under-regulated and culturally unsettled, opening up new artistic possibilities. In this experimental phase, artists explore the new possibilities and resistances of the particular technology and how it can be used artistically. In the next stage, artists appropriate the technology and make it the subject of the artwork itself, revealing its possibilities, limitations and impact on the individual and society. Finally, when the technology is no longer new, either to art or to society, it ceases to be an artistic subject and becomes just another tool in the artist’s toolbox, working silently in the background. In most cases,

artists then go in search of new technologies. The artist interviewed by Ploin et al. (2022, p. 18), Sofia Crespo, for example, notes: “*When a technology becomes very naturalized in art, it stops being exciting. Then we look for the next thing*”. This, in turn, raises further questions about how current artistic engagements with AI relate to this trajectory, whether AI art might be undergoing a parallel transformation. Will the term AI art soon become obsolete, replaced by many different artistic approaches that use AI in one way or another as a mere tool and distance themselves from the terms AI art and AI artist, or will AI art take a new direction?

RQ3: *How these socio-technical changes affect the art world and its logics?*

Since the 17th century, it is part of the art world’s self-image that it is primarily concerned with the creation of the new (Pfisterer, 2011), i.e. that art is oriented towards the regime of the new (Reckwitz, 2012), whether as the “absolute” new, the new through a break with the past, the new through the negation and deconstruction of the “myth of the new”, or the relative new as a recombination of the past or as a revaluation of the given profane (Groys, 1992). Accordingly, art history is confronted with a stream of novelties and innovations (Hutter, 2021). In recent decades, however, the imperative to ‘be innovative’ has extended beyond the arts to many areas of society, establishing itself as the ultimate goal not only of the economy but of contemporary society in general, and as the guiding category of modernity (Göttlich & Kurt, 2012). Given the rapid pace of innovation around AI, the imperative to innovate seems to be taking on unprecedented speed and scale. Accordingly, the question arises as to what the implications of this trend might be for the art field and its regime of creating novelty. Is it conceivable that the very place considered to be the epitome of novelty, namely the arts, might respond to this ‘spiral of innovation’ with rejection and a critical stance towards the imperative of novelty? As media theorist McKenzie Wark (2013) argues, the role of the avant-garde may no longer be to play and subvert, disrupt and innovate, as other fields, especially the dominant ruling class, have long since taken over this role. Perhaps these rapid socio-technical changes around generative AI call for a different kind of avant-garde with the art world.

AI Disclaimer Statement

We, Mariya Dzhimova and Benedikt Zönnchen, acknowledge that we used generative AI—specifically ChatGPT-5 in the preparation of our chapter From Experimental Technology to Mainstream Product. ChatGPT-5 was used for linguistic purposes, namely to refine the language by condensing and shortening passages and by improving stylistic formulation. All conceptual, analytical,

and argumentative content originates from the authors. The generated outputs were reviewed, revised, and approved by the authors before inclusion.

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